

# Chapter 13

## Analytic Proof of the Prime Number Theorem

**Exercise 13.1.** Chebyshev proved that if  $\psi(x)/x$  tends to a limit as  $x \rightarrow \infty$  then this limit equals 1. A proof was outlined in [Exercise 4.26](#). This exercise outlines another proof based on the identity

$$(26) \quad -\frac{\zeta'(s)}{\zeta(s)} = s \int_1^\infty \frac{\psi(x)}{x^{s+1}} dx, \quad (\sigma > 1)$$

given in [Exercise 11.1 \(d\)](#).

(a) Prove that  $(1-s)\zeta'(s)/\zeta(s) \rightarrow 1$  as  $s \rightarrow 1$ .

(b) Let  $\delta = \limsup_{x \rightarrow \infty} (\psi(x)/x)$ . Given  $\varepsilon > 0$ , choose  $N = N(\varepsilon)$  so that  $x \geq N$  implies

$\psi(x) \leq (\delta + \varepsilon)x$ . Keep  $s$  real,  $1 < s \leq 2$ , split the integral (26) into two parts,  $\int_1^N + \int_N^\infty$  and estimate each part to obtain the inequality

$$-\frac{\zeta'(s)}{\zeta(s)} \leq C(\varepsilon) + \frac{s(\delta + \varepsilon)}{s-1},$$

where  $C(\varepsilon)$  is a constant independent of  $s$ . Use (a) to deduce that  $\delta \geq 1$ .

(c) Let  $\gamma = \liminf_{x \rightarrow \infty} (\psi(x)/x)$  and use a similar argument to deduce that  $\gamma \leq 1$ . Therefore if  $\psi(x)/x$  tends to a limit as  $x \rightarrow \infty$  then  $\gamma = \delta = 1$ .

*Proof.* By Theorem 4.9, there are positive constants  $c_1, c_2$  such that  $c_2x \leq \psi(x) \leq c_1x$  eventually holds for all  $x$ . Thus both the liminf and limsup of  $\psi(x)/x$  exist.

(a) By Theorem 12.4  $\zeta(s)$  has a simple pole at  $s = 1$  with residue 1. Therefore there is an entire function  $R(s)$  such that

$$\zeta(s) = \frac{1}{s-1} + R(s) \quad \text{and} \quad \zeta'(s) = -\frac{1}{(s-1)^2} + R'(s).$$

This means

$$\frac{(1-s)\zeta'(s)}{\zeta(s)} = \frac{1 - (s-1)^2 R'(s)}{1 + (s-1)R(s)},$$

hence  $(1-s)\zeta'(s)/\zeta(s) \rightarrow 1$  as  $s \rightarrow 1$ .

(b) Let  $\delta = \limsup \psi(x)/x$  and  $\varepsilon > 0$ . By definition of limsup, there exists  $N = N_\varepsilon$  such that for all  $x \geq N$ ,  $\psi(x) \leq (\delta + \varepsilon)x$ . Then

$$\begin{aligned} -\frac{\zeta'(s)}{\zeta(s)} &= s \left( \int_1^N + \int_N^\infty \right) \frac{\psi(x)}{x^{s+1}} dx \\ &\leq s \int_1^N \frac{\psi(N)}{x^{s+1}} dx + s \int_N^\infty \frac{\delta + \varepsilon}{x^s} dx \\ &= (1 - N^{1-s})\psi(N) + N^{1-s} \frac{s(\delta + \varepsilon)}{s - 1} \\ &\leq \psi(N) + \frac{s(\delta + \varepsilon)}{s - 1}. \end{aligned}$$

Multiplying through by  $s - 1$  gives

$$\frac{(1 - s)\zeta'(s)}{\zeta(s)} \leq (s - 1)\psi(N) + s(\delta + \varepsilon)$$

and applying (a) while taking  $s \rightarrow 1^+$  shows

$$1 \leq \delta + \varepsilon.$$

Letting  $\varepsilon \rightarrow 0^+$  proves  $1 \leq \delta$ .

(c) Let  $\gamma = \liminf \psi(x)/x$  and  $\varepsilon > 0$ . By definition of limsup, there exists  $N = N_\varepsilon$  such that for all  $x \geq N$ ,  $\psi(x) \geq (\gamma - \varepsilon)x$ . Then

$$\begin{aligned} -\frac{\zeta'(s)}{\zeta(s)} &= s \left( \int_1^N + \int_N^\infty \right) \frac{\psi(x)}{x^{s+1}} dx \\ &\geq s \int_1^N \frac{1}{x^{s+1}} dx + s \int_N^\infty \frac{\gamma - \varepsilon}{x^s} dx \\ &= (1 - N^{1-s}) + N^{1-s} \frac{s(\gamma - \varepsilon)}{s - 1} \\ &\geq 1 + \frac{s(\gamma - \varepsilon)}{s - 1}. \end{aligned}$$

Multiplying through by  $s - 1$  gives

$$\frac{(1 - s)\zeta'(s)}{\zeta(s)} \geq (s - 1) + s(\gamma - \varepsilon)$$

and applying (a) while taking  $s \rightarrow 1^+$  shows

$$1 \geq \gamma - \varepsilon.$$

Letting  $\varepsilon \rightarrow 0^+$  proves  $1 \geq \gamma$ . □

**Exercise 13.2.** Let  $A(x) = \sum_{n \leq x} a(n)$ , where

$$a(n) = \begin{cases} 0 & \text{if } n \text{ is not a prime power,} \\ \frac{1}{k} & \text{if } n = p^k. \end{cases}$$

Prove that  $A(x) = \pi(x) + O(\sqrt{x} \log \log x)$ .

*Proof.* Observe by the prime number theorem

$$\begin{aligned} A(x) &= \sum_{n=1}^{\lfloor \log_2(x) \rfloor} \frac{\pi(x^{1/n})}{n} \\ &= \pi(x) + \frac{1}{\log x} \sum_{n=2}^{\lfloor \log_2(x) \rfloor} (x^{1/n} + o(x^{1/n})) \\ &= \pi(x) + \frac{1}{\log x} O(\sqrt{x} \log_2(x)) \\ &= \pi(x) + O(\sqrt{x}). \end{aligned}$$

□

**Exercise 13.3.**

(a) If  $c > 1$  and  $x \neq$  integer, prove that if  $x > 1$ ,

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} \log \zeta(s) \frac{x^s}{s} ds = \pi(x) + \frac{1}{2}\pi(x^{1/2}) + \frac{1}{3}\pi(x^{1/3}) + \cdots.$$

(b) Show that the prime number theorem is equivalent to the asymptotic relation

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} \log \zeta(s) \frac{x^s}{s} ds \sim \frac{x}{\log x} \quad \text{as } x \rightarrow \infty.$$

A proof of the prime number theorem based on this relation was given by Landau in 1903.

*Proof.*

(a) Let  $\sigma = c$ . It was shown in Theorem 11.14 that

$$\log \zeta(s) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{\log n} n^{-s},$$

so by Theorem 11.18 (Perron's formula)

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} \log \zeta(s) \frac{x^s}{s} ds = \sum_{n \leq x} \frac{\Lambda(n)}{\log n}.$$

Observe

$$\frac{\Lambda(n)}{\log n} = \begin{cases} 0 & \text{if } n \text{ is not a prime power,} \\ \frac{1}{k} & \text{if } n = p^k, \end{cases}$$

so

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} \log \zeta(s) \frac{x^s}{s} ds = \sum_{p^k \leq x} \frac{1}{k} = \pi(x) + \frac{1}{2}\pi(x^{1/2}) + \frac{1}{3}\pi(x^{1/3}) + \cdots.$$

(b) By [Exercise 13.2](#) and (a) we have

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} \log \zeta(s) \frac{x^s}{s} ds = A(x) = \pi(x) + O(\sqrt{x} \log \log x),$$

and the proof follows since

$$O(\sqrt{x} \log \log x) = o\left(\frac{x}{\log x}\right).$$

□

**Exercise 13.4.** Let  $M(x) = \sum_{n \leq x} \mu(n)$ . The exact order of magnitude of  $M(x)$  for large  $x$  is not known. In Chapter 4 it was shown that the prime number theorem is equivalent to the relation  $M(x) = o(x)$  as  $x \rightarrow \infty$ . This exercise relates the order of magnitude of  $M(x)$  with the Riemann hypothesis.

Suppose there is a positive constant  $\theta$  such that

$$M(x) = O(x^\theta) \quad \text{for } x \geq 1.$$

Prove that the formula

$$\frac{1}{\zeta(s)} = s \int_1^\infty \frac{M(x)}{x^{s+1}} dx,$$

which holds for  $\sigma > 1$  (see [Exercise 11.1 \(c\)](#)) would also be valid for  $\sigma > \theta$ . Deduce that  $\zeta(s) \neq 0$  for  $\sigma > \theta$ . In particular, this shows that the relation  $M(x) = O(x^{1/2+\varepsilon})$  for every  $\varepsilon > 0$  implies the Riemann hypothesis. It can also be shown that the Riemann hypothesis implies  $M(x) = O(x^{1/2+\varepsilon})$  for every  $\varepsilon > 0$ .

*Proof.* Suppose there is some positive constant  $\theta$  such that  $M(x) = O(x^\theta)$ . Then it's clear

$$s \int_1^\infty \frac{M(x)}{x^{s+1}} dx$$

converges for  $\sigma > \theta$ . As mentioned in [Exercise 11.16](#), the integral is thus analytic in this half-plane. Hence by the uniqueness of analytic continuation,

$$\frac{1}{\zeta(s)} = s \int_1^\infty \frac{M(x)}{x^{s+1}} dx \quad \text{for } \sigma > \theta.$$

Therefore  $1/\zeta(s)$  is analytic in this half-plane, so  $\zeta(s) \neq 0$  for  $\sigma > \theta$ . □

**Exercise 13.5.** Prove the following lemma, which is similar to Lemma 2. Let

$$A_1(x) = \int_1^x \frac{A(u)}{u} du,$$

where  $A(u)$  is a nonnegative increasing function for  $u \geq 1$ . If we have the asymptotic formula

$$A_1(x) \sim Lx^c \quad \text{as } x \rightarrow \infty,$$

for some  $c > 0$  and  $L > 0$ , then we also have

$$A(x) \sim cLx^c \quad \text{as } x \rightarrow \infty.$$

*Proof.* Since  $x^c \rightarrow \infty$  as  $x \rightarrow \infty$ , we can apply L'Hôpital's rule to find

$$L = \lim_{x \rightarrow \infty} \frac{A_1(x)}{x^c} = \lim_{x \rightarrow \infty} \frac{A(x)/x}{cx^{c-1}} = \lim_{x \rightarrow \infty} \frac{A(x)}{cx^c}.$$

□

**Exercise 13.6.** Prove that

$$\frac{1}{2\pi i} \int_{2-\infty i}^{2+\infty i} \frac{y^s}{s^2} ds = 0 \quad \text{if } 0 < y < 1.$$

What is the value of this integral if  $y \geq 1$ ?

*Proof.* When  $0 < y < 1$  consider the contour  $C_1$  illustrated in Figure 5 (a). Since  $y^s/s^2$  is analytic for  $s \neq 0$ ,

$$\frac{1}{2\pi i} \int_{C_1} \frac{y^s}{s^2} ds = 0.$$

Now observe  $|y^s| = y^\sigma \leq y^2$  for any  $s$  on  $C_1$ . Hence if  $C_R$  is the circular sector of  $C_1$ ,

$$\left| \int_{C_R} \frac{y^s}{s^2} ds \right| \leq \frac{y^2}{R^2} (\pi R) \rightarrow 0 \quad \text{as } R \rightarrow \infty,$$

which means

$$\frac{1}{2\pi i} \int_{2-\infty i}^{2+\infty i} \frac{y^s}{s^2} ds = 0.$$

When  $y \geq 1$  consider the contour  $C_2$  illustrated in Figure 5 (b). Note

$$\operatorname{Res}_{s=0} \left( \frac{y^s}{s^2} \right) = \log y,$$

hence

$$\frac{1}{2\pi i} \int_{C_2} \frac{y^s}{s^2} ds = \log y.$$

Now observe  $|y^s| = y^\sigma \leq y^2$  for any  $s$  on  $C_2$ . Hence if  $C_R$  is the circular sector of  $C_2$ ,

$$\left| \int_{C_R} \frac{y^s}{s^2} ds \right| \leq \frac{y^2}{R^2} (2\pi R) \rightarrow 0 \quad \text{as } R \rightarrow \infty,$$

which means

$$\frac{1}{2\pi i} \int_{2-\infty i}^{2+\infty i} \frac{y^s}{s^2} ds = \log y.$$

□

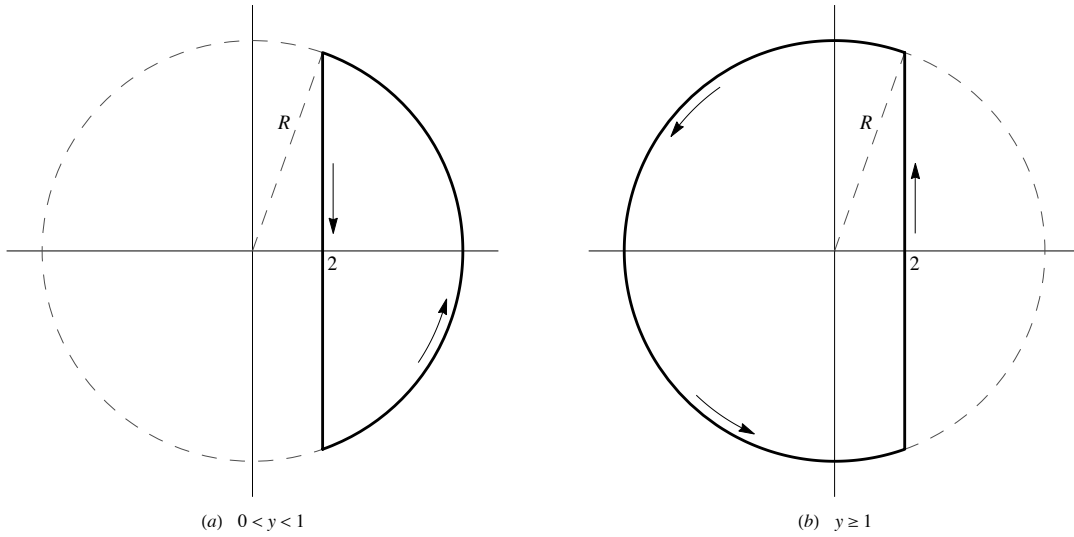


Figure 5: Contours used in Exercise 13.6.

**Exercise 13.7.** Express

$$\frac{1}{2\pi i} \int_{2-\infty i}^{2+\infty i} \frac{x^s}{s^2} \left( -\frac{\zeta'(s)}{\zeta(s)} \right) ds$$

as a finite sum involving  $\Lambda(n)$ .

**Lemma 13.7.** Suppose  $f(s) = \sum_{n=1}^{\infty} a_n n^{-s}$  converges absolutely for  $\sigma > \sigma_a$ . If  $c > \max\{0, \sigma_a\}$ ,

$$\frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} f(s) \frac{x^s}{s^2} ds = \sum_{n \leq x} a_n \log \left( \frac{x}{n} \right).$$

*Proof of Lemma.* Applying Exercise 13.6, for  $c > \max\{0, \sigma_a\}$ ,

$$\begin{aligned} \frac{1}{2\pi i} \int_{c-\infty i}^{c+\infty i} f(s) \frac{x^s}{s^2} ds &= \sum_{n=1}^{\infty} \frac{a_n}{2\pi i} \int_{c-\infty i}^{c+\infty i} \frac{(x/n)^s}{s^2} ds \\ &= \sum_{n \leq x} a_n \log \left( \frac{x}{n} \right). \end{aligned}$$

□

*Proof of Exercise.* Let

$$f(s) = -\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

By Lemma 13.7,

$$\frac{1}{2\pi i} \int_{2-\infty i}^{2+\infty i} \frac{x^s}{s^2} \left( -\frac{\zeta'(s)}{\zeta(s)} \right) ds = \sum_{n \leq x} \Lambda(n) \log \left( \frac{x}{n} \right).$$

□

**Exercise 13.8.** Let  $\chi$  be any Dirichlet character mod  $k$  with  $\chi_1$  the principal character. Define

$$F(\sigma, t) = 3 \frac{L'}{L}(\sigma, \chi_1) + 4 \frac{L'}{L}(\sigma + it, \chi) + \frac{L'}{L}(\sigma + 2it, \chi^2).$$

If  $\sigma > 1$  prove that  $F(\sigma, t)$  has real part equal to

$$-\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma}} \operatorname{Re} \{3\chi_1(n) + 4\chi(n)n^{-it} + \chi^2(n)n^{-2it}\}$$

and deduce that  $\operatorname{Re} F(\sigma, t) \leq 0$ .

*Proof.* Let  $\sigma > 1$ . By Theorem 11.14,  $L(s, \chi) = e^{G(s)}$  where

$$G(s) = \sum_{n=2}^{\infty} \frac{\Lambda(n)\chi(n)}{\log n} n^{-s}.$$

Differentiating gives

$$L'(s, \chi) = G'(s)e^{G(s)} = G'(s)L(s, \chi),$$

therefore

$$\begin{aligned} F(\sigma, t) &= -3 \sum_{n=1}^{\infty} \frac{\Lambda(n)\chi_1(n)}{n^{\sigma}} - 4 \sum_{n=1}^{\infty} \frac{\Lambda(n)\chi(n)}{n^{\sigma+it}} - \sum_{n=1}^{\infty} \frac{\Lambda(n)\chi^2(n)}{n^{\sigma+2it}} \\ &= - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma}} \{3\chi_1(n) + 4\chi(n)n^{-it} + \chi^2(n)n^{-2it}\}. \end{aligned}$$

Taking the real part shows

$$\operatorname{Re} F(\sigma, t) = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^{\sigma}} \operatorname{Re} \{3\chi_1(n) + 4\chi(n)n^{-it} + \chi^2(n)n^{-2it}\}.$$

Letting  $\chi(n) = e^{i\theta}$  gives

$$\operatorname{Re} \{4\chi(n)n^{-it}\} = 4 \cos(\theta) \cos(t \log n) + 4 \sin(\theta) \sin(t \log n) = 4 \cos(\theta - t \log n)$$

and similarly

$$\operatorname{Re} \{\chi^2(n)n^{-2it}\} = \cos(2(\theta - t \log n)).$$

Thus if  $(n, k) = 1$ , then

$$\begin{aligned} \operatorname{Re} \{3\chi_1(n) + 4\chi(n)n^{-it} + \chi^2(n)n^{-2it}\} &= 3 + 4 \cos(\theta - t \log n) + \cos(2(\theta - t \log n)) \\ &= 2(1 + \cos(\theta - t \log n))^2 \geq 0. \end{aligned}$$

If  $(n, k) \neq 1$  then

$$3\chi_1(n) + 4\chi(n)n^{-it} + \chi^2(n)n^{-2it} = 0,$$

so it follows that  $\operatorname{Re} F(\sigma, t) \leq 0$ . □

**Exercise 13.9.** Assume that  $L(s, \chi)$  has a zero of order  $m \geq 1$  at  $s = 1 + it$ . Prove that for this  $t$  we have:

(a)  $\frac{L'}{L}(\sigma + it, \chi) = \frac{m}{\sigma - 1} + O(1)$  as  $\sigma \rightarrow 1^+$ , and

(b) there exists an integer  $r \geq 0$  such that

$$\frac{L'}{L}(\sigma + 2it, \chi^2) = \frac{r}{\sigma - 1} + O(1) \quad \text{as } \sigma \rightarrow 1^+,$$

except when  $\chi^2 = \chi_1$  and  $t = 0$ .

*Proof.*

(a) We have  $L(\sigma + it, \chi) = (\sigma - 1)^m R(\sigma + it)$ , where  $R(\sigma + it) \neq 0$  in a small neighborhood about  $1 + it$ . Logarithmically differentiating shows

$$\frac{L'(\sigma + it, \chi)}{L(\sigma + it, \chi)} = \frac{m}{\sigma - 1} + \frac{R'(\sigma + it)}{R(\sigma + it)}.$$

Since  $R(\sigma + it) \neq 0$  for  $\sigma$  near 1,  $R'(\sigma + it)/R(\sigma + it)$  must be bounded as  $\sigma \rightarrow 1^+$ .

(b) Suppose  $\chi^2 \neq \chi_1$  or  $t \neq 0$ , i.e.  $L(s, \chi^2)$  is analytic at  $s = 1 + 2it$ . Define  $r \geq 0$  to be the order of the zero of  $L(s, \chi^2)$  at  $s = 1 + it$ . (Note if there is no zero, then  $r = 0$ .) Mimicking (a) shows

$$\frac{L'(\sigma + 2it, \chi^2)}{L(\sigma + 2it, \chi^2)} = \frac{r}{\sigma - 1} + O(1) \quad \text{as } \sigma \rightarrow 1^+.$$

□

**Exercise 13.10.** Use Exercises 8 and 9 to prove that

$$L(1 + it, \chi) \neq 0 \quad \text{for all real } t \text{ if } \chi^2 \neq \chi_1$$

and that

$$L(1 + it, \chi) \neq 0 \quad \text{for all real } t \neq 0 \text{ if } \chi^2 = \chi_1.$$

[Hint: Consider  $F(\sigma, t)$  as  $\sigma \rightarrow 1^+$ .]

*Proof.* Recall for  $\sigma > 1$ ,

$$L(\sigma, \chi_1) = \zeta(\sigma) \prod_{p|k} (1 - p^{-\sigma}) = c_k(\sigma) \zeta(\sigma).$$

Thus  $L(\sigma, \chi_1)$  has a simple pole at  $\sigma = 1$  with residue  $c_k(1)$ , and long division shows

$$3 \frac{L'(\sigma, \chi_1)}{L(\sigma, \chi_1)} = \frac{3}{\sigma - 1} + O(1) \quad \text{as } \sigma \rightarrow 1^+.$$

Suppose  $L(1 + it, \chi) = 0$ . If  $\chi^2 \neq \chi_1$  or  $t \neq 0$ , then by [Exercise 13.9](#),

$$F(\sigma, t) = \frac{3 + 4m + r}{\sigma - 1} \quad \text{as } \sigma \rightarrow 1^+.$$

Hence since  $3 + 4m + r > 0$ ,  $F(\sigma, t) \rightarrow \infty$  as  $\sigma \rightarrow 1^+$ . This contradicts [Exercise 13.8](#), which says  $\operatorname{Re} F(\sigma, t) \leq 0$ . This means  $L(1 + it, \chi) \neq 0$ . □



**Exercise 13.11.** For any arithmetical function  $f(n)$ , prove that the following statements are equivalent:

- (a)  $f(n) = O(n^\varepsilon)$  for every  $\varepsilon > 0$  and all  $n \geq n_1$ .
- (b)  $f(n) = o(n^\delta)$  for every  $\delta > 0$  as  $n \rightarrow \infty$ .

*Proof.* If  $f(n) = O(n^\varepsilon)$  for every  $\varepsilon > 0$  and all  $n \geq n_1$ , then for all  $\delta > \varepsilon$

$$f(n) = O(n^\varepsilon) = o(n^{\varepsilon+\delta}).$$

This shows (a) implies (b), since  $\varepsilon + \delta$  can be as close to 0 as we like.

If  $f(n) = o(n^\delta)$  for every  $\delta > 0$ , then in particular  $f(n) = o(n)$ . Hence there exists  $n_1$  such that for all  $n \geq n_1$ ,  $|f(n)| \leq n$ . This means for any  $\varepsilon \geq 1$ ,

$$f(n) = O(n^\varepsilon) \quad \text{for } n \geq n_1.$$

Now for  $0 < \varepsilon < 1$  there exists  $n_\varepsilon$  such that for all  $n \geq n_\varepsilon$ ,  $|f(n)| \leq n^\varepsilon$ . Let

$$m_\varepsilon = \max_{n_1 \leq n \leq n_\varepsilon} \left| \frac{f(n)}{n^\varepsilon} \right| \quad \text{and} \quad M_\varepsilon = \max\{m_\varepsilon, 1\}.$$

Then for all  $n \geq n_1$  we have  $|f(n)| \leq M_\varepsilon n^\varepsilon$ , i.e.  $f(n) = O(n^\varepsilon)$  and all  $n \geq n_1$ . □

**Exercise 13.12.** Let  $f(n)$  be a multiplicative function such that if  $p$  is prime then

$$f(p^m) \rightarrow 0 \quad \text{as } p^m \rightarrow \infty.$$

That is, for every  $\varepsilon > 0$  there is an  $N(\varepsilon)$  such that  $|f(p^m)| < \varepsilon$  whenever  $p^m > N(\varepsilon)$ . Prove that  $f(n) \rightarrow 0$  as  $n \rightarrow \infty$ .

[Hint: There is a constant  $A > 0$  such that  $|f(p^m)| < A$  for all primes  $p$  and all  $m \geq 0$ , and a constant  $B > 0$  such that  $|f(p^m)| < 1$  whenever  $p^m > B$ .]

*Proof.* Following the hint let  $M$  be the number of prime powers  $\leq B$ , which means  $A^M$  is a fixed constant. Letting  $n = p_1^{\alpha_1} \cdots p_s^{\alpha_s} \cdot q_1^{\beta_1} \cdots q_t^{\beta_t}$  where  $p_i^{\alpha_i} \leq B$  and  $q_j^{\beta_j} > B$ , then

$$|f(n)| \leq A^M \prod_{i=1}^t |f(q_i^{\beta_i})|.$$

Now  $n$  can approach  $\infty$  in two ways.

- If  $\beta_i \rightarrow \infty$  as  $n \rightarrow \infty$  for some  $i$ , then  $f(q_i^{\beta_i}) \rightarrow 0$  which means  $f(n) \rightarrow 0$  too.
- If  $n \rightarrow \infty$  but  $\beta_i \not\rightarrow \infty$  for all  $i$ , it must be that  $t \rightarrow \infty$ . Assuming  $q_1^{\beta_1} < q_2^{\beta_2} < \cdots < q_t^{\beta_t}$ , then  $q_t \rightarrow \infty$  as  $t \rightarrow \infty$ . Thus for any  $\varepsilon > 0$  there is a  $t$  large enough such that  $|f(q_t^{\beta_t})| < \varepsilon$ . Hence

$$|f(n)| \leq A^M \prod_{i=1}^t |f(q_i^{\beta_i})| < A^M \varepsilon,$$

or in other words  $f(n) \rightarrow 0$  as  $n \rightarrow \infty$ . □

**Exercise 13.13.** If  $\alpha \geq 0$  let  $\sigma_\alpha(n) = \sum_{d|n} d^\alpha$ . Prove that for every  $\delta > 0$  we have

$$\sigma_\alpha(n) = o(n^{\alpha+\delta}) \quad \text{as } n \rightarrow \infty.$$

[*Hint:* Use [Exercise 13.12](#).]

*Proof.* Fix  $\delta > 0$  and define  $f(n) = \sigma_\alpha(n)/n^{\alpha+\delta}$ . Observe  $f(n)$  is multiplicative and

$$f(p^m) = \frac{1}{p^{m\alpha+m\delta}} \frac{p^{(m+1)\alpha} - 1}{p^\alpha - 1} = \frac{1}{(p^m)^\delta} \left\{ \frac{p^\alpha - p^{-m\alpha}}{p^\alpha - 1} \right\},$$

hence  $f(p^m) \rightarrow 0$  as  $p^m \rightarrow \infty$ . So applying [Exercise 13.12](#) it's clear  $f(n) = o(1)$ , i.e.  $\sigma_\alpha(n) = o(n^{\alpha+\delta})$ .  $\square$